

Panagiotis Aigaias Dimtsoudis

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Education

University of Pennsylvania

Bachelor of Science in Electrical Engineering | **GPA: 3.68/4.00**

Philadelphia, PA

Expected May 2028

Relevant Coursework: Fundamentals of Solid-State Circuits, Embedded Systems Laboratory, Signal and Information Processing, Electronic, Photonic, and Electromechanical Devices, Principles of Optics and Photonics, Electrical Circuits and Systems

Project Portfolio

Metal Detector PCB

- Designed and laid out an analog metal detector PCB with ALD1103 MOSFETs, LC oscillators, mixer, and filters.
- Verified oscillator frequencies, bias points, gain, and output via hand calculations and LTspice simulations.
- Assembled, soldered, and tested the PCB, using an oscilloscope to debug each stage and drive an 8Ω speaker.

Wireless ECG Monitor (Analog Front-End + BLE)

- Built a portable ECG front-end using an instrumentation amp, virtual ground biasing, and band-limited filtering.
- Sampled ECG at ~200 Hz on a Feather MCU and streamed it over BLE for live plotting on a central device.
- Wrote Python scripts for logging, plotting, and R-peak detection to estimate heart rate (BPM).

Guitar Hero Synthesizer

- Repurposed a Guitar Hero controller into a playable embedded synthesizer using an ATmega328PB.
- Wrote bare-metal C firmware for real-time fret, strum, mute, joystick, whammy, and LCD control.
- Implemented PWM audio synthesis with power-chord playback, vibrato, and decaying note behavior.
- Debugged power and audio issues by redesigning the speaker and power regulation path.

Mesh Network for Hiking Trails

- Optimized Arduino/XBee mesh placement for ~100–150 m hops and multi-mile trail coverage.
- Implemented info/distress modes with routing and ACKs for reliable message delivery.
- Built a low-cost prototype and test plan for range, packet success rate, and power draw.

Experience

Device Research and Engineering Laboratory

Philadelphia, PA

Research Assistant

May 2026 – Present

- Contributed to an IEDM 2026 submission on ultrathin AlN/AlScN/AlN ferroelectric junctions for cryogenic memory.
- Performed cryogenic electrical testing (DC/AC J–V, C–V, PUND) on 6 nm AlScN devices from 400 K to 14 K.
- Gathered and analyzed switching, retention, and endurance data to characterize device reliability trends.
- Built 3D CAD models of the device stack and crossbar array to support compute-in-memory design.
- Wrote Python scripts to process, analyze, and plot measurement data (J–V, C–V, PUND) to identify device trends.

Technology Support Services

Philadelphia, PA

Technology Support Specialist

Jan 2025 – Present

- Provide software/hardware support to Penn undergraduates through in-person help and ticketing.
- Triage technical issues, document cases, coordinate escalations, and support equipment loaner operations.

Automation Systems Hellas

Thessaloniki, Greece

Computer Engineering Intern

Jun 2024 – Aug 2024

- Updated control panel documentation by labeling field wiring and tracking I/O device inventories.
- Supported OT and IT tasks by imaging PCs and HMIs and monitoring servers and network devices.
- Wrote C and Python utilities for log parsing and configuration checks to streamline commissioning.

Academics & Activities

Science and Technology Wing (STWing)

Philadelphia, PA

Member

Sep 2024 – Present

- Built and debugged drone/blimp prototypes through subsystem wiring, bench testing, and flight validation.

Robotics Club / F1 in Schools

Thessaloniki, Greece

IT Lead

Sep 2021 – May 2024

- Led website/outreach and contributed to CAD design/testing for a nationally qualifying F1 team.

Skills

Programming & Embedded Systems: Python, Java, OCaml, C/C++, Assembly, UART/I2C, oscilloscopes, soldering

CAD/EDA: Altium Designer, LTspice, SolidWorks, Fusion 360

Interests: Embedded systems, PCB/chip layout, analog circuits, nanofabrication, ferroelectric memory

Languages: Greek (Native), English (Fluent), German (Intermediate), Japanese (Intermediate)